

Machine learning methods for continuous glucose monitoring data and their potential in cardiovascular risk assessment

PhD dissertation

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Disclosures and Notes

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Notes on website version

This dissertation is available for viewing online at HeleneThomsen.github.io/dissertation/ or by scanning the QR code in the lower right-hand corner of the title page.

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Preface

Outline of the dissertation

Papers in the dissertation

Study I

Thomsen, H. B., Lebiecka-Johansen, B., Nørgaard, O., Andersen, T. H., Andersen, S. T., Fagherazzi, G., Hulman, A., & Isaksen, A. A. (2026). Continuous Glucose Monitoring–Derived Metrics and Cardiovascular Risk Among People With Diabetes: Systematic Scoping Review. *JMIR Diabetes*, 11(1), e89374. <https://doi.org/10.2196/89374>

Study II

Thomsen, H. B., Andersen, S. T., Fagherazzi, G., Isaksen, A. A. & Hulman, A. (2026). Foundation model–derived representations of continuous glucose monitoring data: relation to glycemic patterns and cardiovascular risk. (Under review at *JMIR AI*, (7.30.2026), pending for preprint at *JMIR preprints*)

Study III

Thomsen, H. B., Li, L. Y., Isaksen, A. A., Lebiecka-Johansen, B., Bour, C., Fagherazzi, G., Doorn, W. P. T. M. van, Varga, T. V., & Hulman, A. (2025). Racial disparities in continuous glucose monitoring-based 60-min glucose predictions among people with type 1 diabetes. *PLOS Digital Health*, 4(6), e0000918. <https://doi.org/10.1371/journal.pdig.0000918>

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Abbreviations

AGP: ambulatory glucose profile

AID: Automated insulin delivery

AUROC: Area Under the Receiver Operating Characteristic Curve

CGM: continuous glucose monitoring

CV: coefficient of variation

CVD: cardiovascular disease

eGFR: Estimated Glomerular Filtration Rate

GMI: glucose management indicator

HbA1c: hemoglobin A1c

MBG: mean glucose

NT-proBNP: N-terminal Pro-B-type Natriuretic Peptide

RMSE: root mean square error

SCORE2-Diabetes: Systematic Coronary Risk Evaluation 2-Diabetes

SMBG: self-monitoring blood glucose

T1D: type 1 diabetes

T2D: type 2 diabetes

TAR: time above range

TBR: time below range

TIR: time in range

1 Introduction

1.1 Diabetes

Diabetes is a metabolic disease characterized by high blood glucose levels known as hyperglycemia.^{1,2} The disease is one of the fastest growing global health challenges.³ In 2021 it was estimated that 537 million people lived with diabetes, and it is expected to reach 643 million by 2030.³ The most common forms of chronic diabetes are type 1 diabetes (T1D) and type 2 diabetes (T2D).¹ T1D is an autoimmune disease characterized by the destruction of the pancreatic islet β -cells, leading to an absolute insulin deficiency.⁴ Approximately 85%-95% of all people with diabetes in developing countries have T2D.¹ It is the most common form of diabetes and in contrast to T1D it generally begins with insulin resistance in which the body's cells become less responsive to insulin.^{1,5} Initially, the pancreas compensates by increasing insulin secretion, but over time progressive β -cell dysfunction results in inadequate insulin production to meet the body's demands.^{1,2,5} T2D is the most common form of diabetes. Despite their distinct underlying pathophysiology, both T1D and T2D are characterized by chronic hyperglycemia.

Diabetes can be diagnosed based on a bloodtest to check hemoglobin A1c (HbA1c) levels or through plasma glucose criteria (Table 1.1).^{4,6}

Although both plasma glucose tests and HbA1c tests are appropriate for diabetes screening, the HbA1c test has several advantages.⁴ Some of the benefits are that it requires no fasting and it is less sensitive to stress and changes in nutrition.⁴ HbA1c is an indirect measure of blood glucose levels because glucose binds hemoglobin over ~ 120 days lifespan of erythrocytes (red blood cells).⁴ However HbA1c testing is not always accessible, affordable or suitable (e.g., for people with anemia or under HIV-treatment).⁴ With the guidelines recommendation of using HbA1c, a debate has sparked on the controversy of using a set thresholds as it is well known that racial differences in HbA1c exists.⁷ Black individuals have higher HbA1c levels than Hispanic and non-Hispanic White people with diabetes,^{4,8} however studies show that the association between HbA1c and risk of complications appear to be similar between ethnic race groups.⁴

1.2 Blood Glucose Monitoring

HbA1c values are a very important part of diabetes management. The treatment goal is for HbA1c to be below 53 mmol/mol for non-pregnant adults without any hyperglycemia or hypoglycemia (low blood glucose levels). However HbA1c goals should be individualized to fit each individual's specific situation. HbA1c can vary from e.g., <48 mmol/mol to none.^{9,10} There are two tools available for self-assessment of glycemia, the self-monitoring blood glucose (SMBG) device measuring capillary blood with finger-sticks, and the continuous glucose monitoring (CGM) measuring interstitial fluid continuously.^{10,11} Both monitoring devices have shown to improve glycemic control in people with diabetes, however the CGM is superior when it comes to immediate information and alerts on asymptomatic hypo- and hyperglycemic events and

Table 1.1: Diagnostic tests for diabetes

Test	Criteria	Description
Fasting plasma glucose	≥ 7.0 [mmol/L]	Fasting is defined as no calorie intake for at least eight hours.
2-hour plasma glucose	≥ 11.1 [mmol/L]	Glucose is measured 2 hours after oral ingestion of 75g glucose dissolved in water (OGTT)
Random plasma glucose	≥ 11.1 [mmol/L]	Individuals with classic symptoms of hyperglycemia (e.g., dehydration and unintentional weight loss) or hyperglycemic crisis (e.g., diabetic ketoacidosis) together with a plasma test taken any time of day.
Glycated hemoglobin (HbA1c)	≥ 48 [mmol/mol]	

provide a comprehensive glucose profile including both pre- and postprandial blood glucose measurements.^{10,12,13}

The CGM devices measures the glucose in the interstitial fluid, which is the fluid around and between cells. It corresponds with plasma glucose although a lag can appear when glucose levels rise or fall rapidly.¹⁴ Use of CGM devices has shown to improve glycemic control in both T1D and T2D with improvements in HbA1c levels and more time spent in target range (3.9-10 mmol/L) regardless of age, sex, education and income.^{9,14} CGM devices focuses on monitoring and it is possible to get an ambulatory glucose profile (AGP) report from you device (Figure 1.1). The AGP report shows summary statistics over all your data (usually a two weeks period) including CGM-derived metrics as shown in (Table 1.2).¹² Besides monitoring, data from CGM devices are now used in combination with insulin delivery systems to automate the process of administering insulin.¹⁴ Automated insulin delivery (AID) systems typically use data from CGM devices to determine or adjust insulin delivery. The simplest AIDs are threshold-based low-glucose suspend systems in which insulin delivery is automatically suspended when blood glucose levels drop below a predefined threshold.¹⁵ More advanced systems utilize additional information such as carbohydrate intake and physical activity to optimize insulin delivery.¹⁷

Glucose Pattern Insights

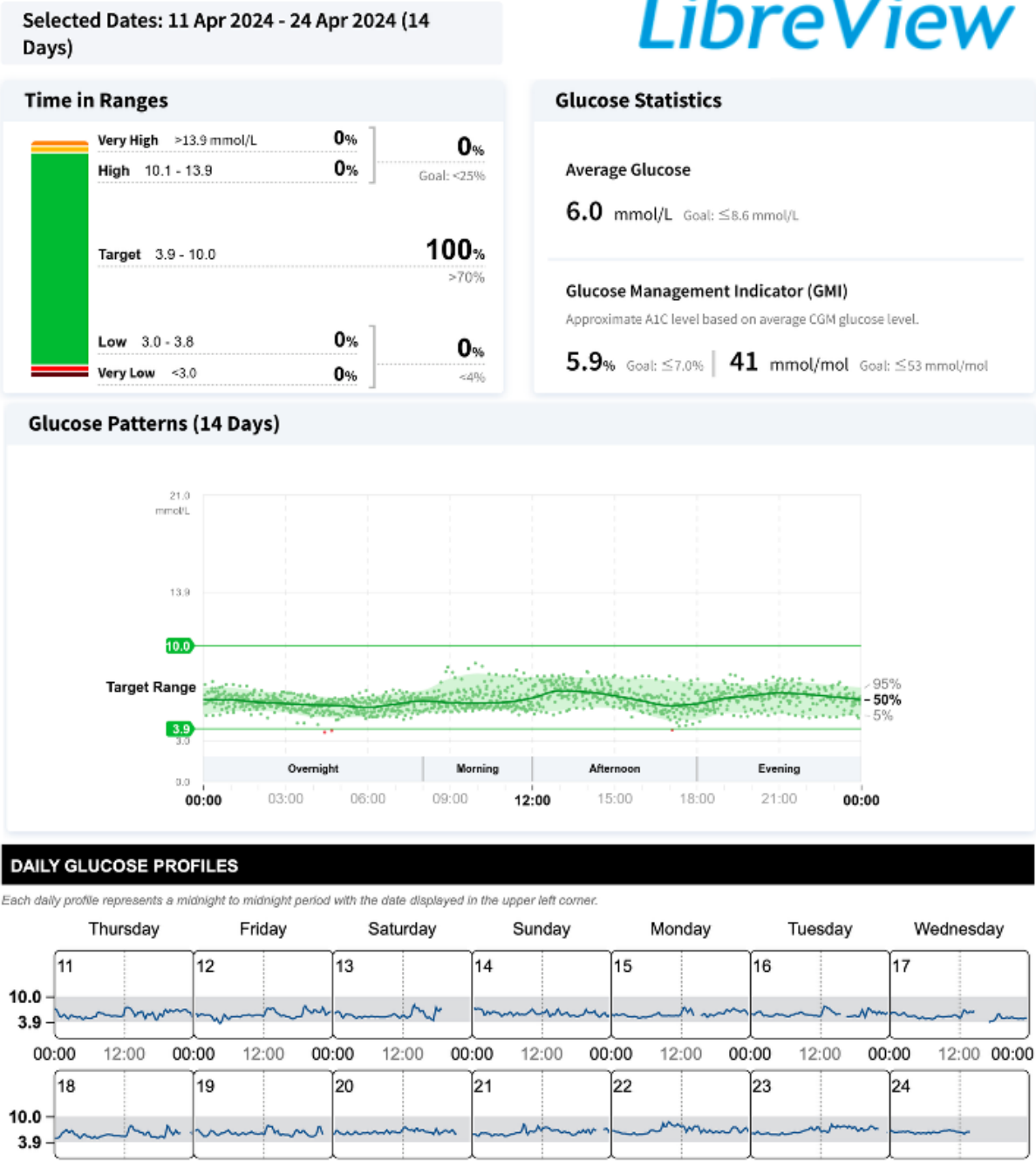


Table 1.2: CGM-derived metrics presented in the ambulatory glucose profile report with target values.

Abbreviation	CGM-derived metric	Range	Recommended target
GMI	Glucose Management Indicator		No specific target
TIR	Time in Range	3.9-10.0 mmol/L (70-180 mg/dL)	> 70%
TBR ₅₄	Time Below Range	< 3.0 mmol/L (< 54 mg/dL)	< 1%
TBR ₇₀	Time Below Range	< 3.9 mmol/L (< 70 mg/dL)	< 4%*
TAR ₁₈₀	Time Above Range	> 10.0 mmol/L (> 180 mg/dL)	< 25%**
TAR ₂₅₀	Time Above Range	> 13.9 mmol/L (> 250 mg/dL)	< 5%
CV	Coefficient of Variation		≤ 36%
	Number of days sensor was worn		≥ 14 days for pattern management
	Active sensor time		70% of 14 days

* This goal include both TBR₅₄ and TBR₇₀.

** This goal include both TAR₁₈₀ and TAR₂₅₀

including both peaks and nadirs.¹⁹ Describing the variability is clinically relevant because two individuals with similar HbA1c levels may exhibit different glycemic variability profiles.¹⁸ Numerous metrics have been developed to characterize different aspects of glucose variability, but no single measure captures all of its dimensions. Consequently, no universal consensus has been reached, although the coefficient of variation (CV) and SD have been incorporated into AGP reports and the American Diabetes Association guidelines (Table 1.2).^{18,19}

1.4 Cardiovascular Diseases and Heart Failure

Although advances in diabetes management have reduced the incidence of several diabetes-related complications, cardiovascular disease remains a major cause of morbidity and mortality in people with diabetes.^{20–22} Chronic hyperglycemia can lead to long-term complications, although the progression and manifestation of these complications vary between individuals.⁴ Identifying individuals at increased risk of atherosclerotic cardiovascular disease or heart failure before clinical disease develops is therefore an important component of diabetes care.²¹

Heart failure, as the condition is at least twice as prevalent among people with diabetes as in the general population.^{21,23} It may present as either an acute or a chronic condition which is characterized by a combination of symptoms, including breathlessness, fatigue, and ankle swelling, together with clinical signs such as elevated jugular venous pressure and pulmonary crackles.²⁴ In individuals with suspected chronic heart failure based on risk factors, symptoms, clinical findings, and supporting investigations such as electrocardiography, measurement of N-terminal Pro-B-type Natriuretic Peptide (NT-proBNP) is recommended as part of the diagnostic work-up.²⁴ An NT-proBNP concentration of 125 pg/mL indicates the need for further cardiac evaluation, whereas concentrations below this threshold make chronic heart failure unlikely and

alternative diagnoses should be considered.²⁴ Consequently, NT-proBNP represents a clinically relevant biomarker for identifying individuals at increased risk of chronic heart failure.

1.5 Clinical Prediction Models

To facilitate early identification of individuals at increased cardiovascular risk, several clinical prediction models have been developed.^{22,25–27} By estimating the probability of a cardiovascular disease (CVD) event within a specified time horizon, these models can increase awareness of cardiovascular risk, encourage adherence to recommended lifestyle changes and medical treatment, and help identify individuals who are most likely to benefit from preventive interventions.²⁸ Among these, the Systematic Coronary Risk Evaluation 2-Diabetes (SCORE2-Diabetes) model estimates the 10-year risk of CVD in individuals with T2D. It was developed and calibrated using European cohorts, making it particularly relevant for the Danish cohort included in this dissertation.²² However, the SCORE2-Diabetes model relies solely on traditional clinical variables, including age, sex, smoking status, systolic blood pressure, high-density lipoprotein cholesterol, age at diabetes diagnosis, HbA1c, and Estimated Glomerular Filtration Rate (eGFR), and does not incorporate information on glucose dynamics beyond the long-term glycemic exposure reflected by HbA1c.²² HbA1c have been independently associated with risk of heart failure,²⁹ however it does not capture the short-term glucose fluctuations, which may be associated with the progression of vascular complications in diabetes.³⁰ However association is not prediction and this distinction is important, as association and prediction are fundamentally different objectives, despite the terms sometimes being used interchangeably in the medical literature.³¹ Association studies aim to identify relationships between variables and an outcome at group level, whereas prediction models are developed to estimate future outcomes for individual patients.³¹

1.6 Association vs Prediction

Traditional statistical modelling is often used to test hypotheses regarding associations between exposures X and outcomes Y .³² Regression models applied to observational data are commonly used to evaluate hypotheses derived from prior knowledge, where the choice of exposures is guided by existing theory.³² These approaches assume that the chosen statistical model adequately represents the underlying data-generating mechanism, allowing conclusions to be drawn from the estimated model parameters.³³ Results from association studies are typically summarized using estimated effect measures, such as regression coefficients, odds ratios, hazard ratios, or risk ratios, together with confidence intervals and significance tests,³¹

In contrast, prediction modelling applies statistical models to predict future outcomes Y based on a set of predictors X .³³ Prediction models may be implemented using a wide range of approaches, including regression models, decision trees, and machine learning algorithms.^{32,33} The prediction models are fitted to data (the predictors) and the output Y' is evaluated based on your outcome Y ,³³ and evaluation metrics on prediction studies often include classification metrics such as Area Under the Receiver Operating Characteristic Curve (AUROC), sensitivity, specificity and calibration plots³⁴ or correlation coefficients, and root mean square error (RMSE), for forecasting models.³⁵

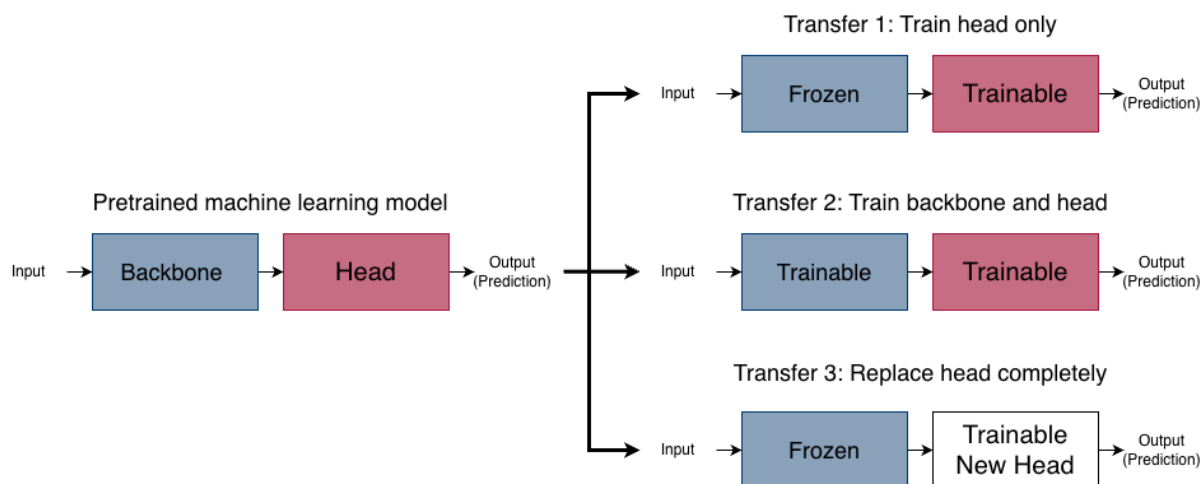
1.7 Machine Learning and Representation Learning

Associations between CGM-derived metrics and CVD outcomes have been investigated in the literature,^{36–38} whereas evidence regarding the predictive value of CGM-derived metrics for cardiovascular risk assessment remains limited.^{39,Hua2012?,Koroleva2022?} The widespread adoption

of CGM has generated large volumes of glucose data, enabling the application of machine learning methods in diabetes care.^{16,40} Machine learning models are often divided into supervised and unsupervised machine learning.⁴¹ Supervised machine learning uses labelled data where each sample (e.g. age and sex from one individual) has a label or an outcome (e.g., elevated NT-proBNP) to predict new outcomes for new unforeseen data. Unsupervised machine learning on the other hand uses data without labels to explore underlying structures or clusters in data.⁴¹ Among supervised learning methods, deep neural networks are particularly notable because they can automatically learn informative representations directly from raw data for downstream tasks such as prediction or classification, reducing the need for manual feature engineering. This process is known as representation learning.^{16,42} Consequently, deep neural networks offer an alternative to relying on predefined CGM-derived metrics. Although international consensus recommendations advocate the routine reporting of glucose variability using CV, MAGE, and SD, with particular emphasis on CV,¹² glucose variability encompasses multiple dimensions, including within- and between-day variability, the amplitude of glucose excursions, and temporal glucose patterns.¹⁸ As a result, no single metric has been shown to comprehensively capture all aspects of glucose variability, and no consensus currently exists regarding the optimal measure.^{18,19}

1.8 Foundation Models and Transfer Learning

With the scale and scope of some of the machine learning models that have developed over the years, a new term has arisen, foundation models.⁴³ They are defined as any models that are trained on a huge amount of data and can be specialized to solve a wide range of different usecases also known as downstream tasks. Two CGM specialized foundation models exist in the literature which both use self-supervised machine learning.^{44,45} Self-supervised machine learning is a technique used to train models on unlabelled data through different techniques, one of them including masked learning (e.g. predicting artificially masked words in text or values in time-series signals).^{43–45} A pretrained foundation model is therefore a model first trained on a task such as predicting masked values in a CGM signal and thereby has learned the representations of blood glucose dynamics measured by a CGM device. The pretrained foundation model can then be used in a different task (e.g., predicting CVD outcomes⁴⁵) using transfer learning.⁴³ Transfer learning is based on the premise that a pretrained foundation model has learned general representations that can be transferred to new, related tasks without training a model from scratch.⁴³ There are multiple different transfer learning strategies as shown in Figure H.0.⁴⁶ One of the main challenges of deep neural networks is their requirement for large amounts of training data. Transfer learning addresses this challenge by reusing knowledge learned from large datasets and applying it to related tasks with limited labelled data. This is particularly relevant in clinical risk prediction, where labelled data are often scarce, especially for minority populations.^{46–48}



Overview of transfer learning strategies. Based on Deng et al. (2021),⁴⁶ under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence. ::git

1.9 Fairness and bias in machine learning

- Define race and ethnicity
- Algorithmic bias and group fairness
- Fairness metrics and mitigation strategies
- Trends in the quality of care and racial disparities in Medicare managed care: <https://pubmed.ncbi.nlm.nih.gov/16107622/>

2 Methods

In this chapter the three studies included in this PhD will be presented. The data will be described together with the chosen methods of the three studies and the different clinical variables, CGM-derived metrics and relevant cardiovascular outcomes will be explained.

- ☐ What was done (Description)
- ☐ Why it was done (Justification)
- ☐ tell about the 7-point fingerprick?

2.1 Overview of the Studies

	Study I	Study II	Study III
Title	Continuous Glucose Monitoring-Derived Metrics and Cardiovascular Risk Among People With Diabetes: Systematic Scoping Review	Foundation model-derived representations of continuous glucose monitoring data: relation to glycemic patterns and cardiovascular risk	Racial disparities in continuous glucose monitoring-based 60-min glucose predictions among people with type 1 diabetes
18			
Study	Review	CGM foundation model	Algorithmic fairness study